

ISSW 2026 · Saturday 3 October · 09:00–16:00 · Peak Room

autoATES v3.0

Mapping Connaught Creek with open-source tools

You can run the notebooks on a laptop, or follow the same maps on the projector.

Two tracks, one timetable

On a laptop

Run the notebook for the current module (Python 3.11 + QGIS). If a step stalls, copy that folder's outputs_reference and continue with the group.

Watching

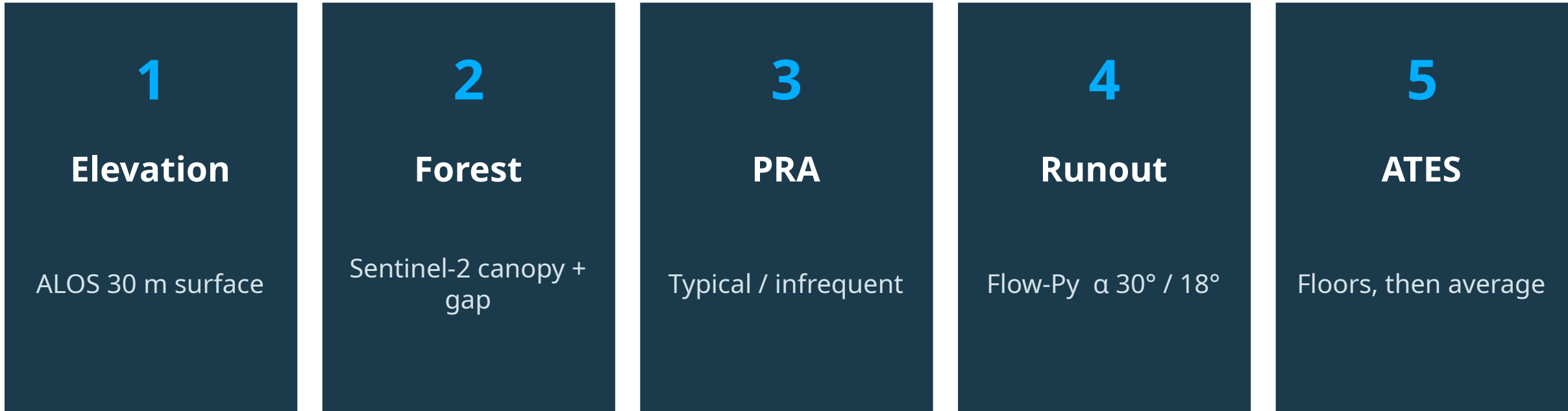
The same maps are on this screen. A laptop is optional. You leave with the method, the design choices, and where to get the code.

If a step stalls

data/<step>/outputs_reference/ lets you stay with the room. Helpers will keep the day moving so the afternoon still fits.

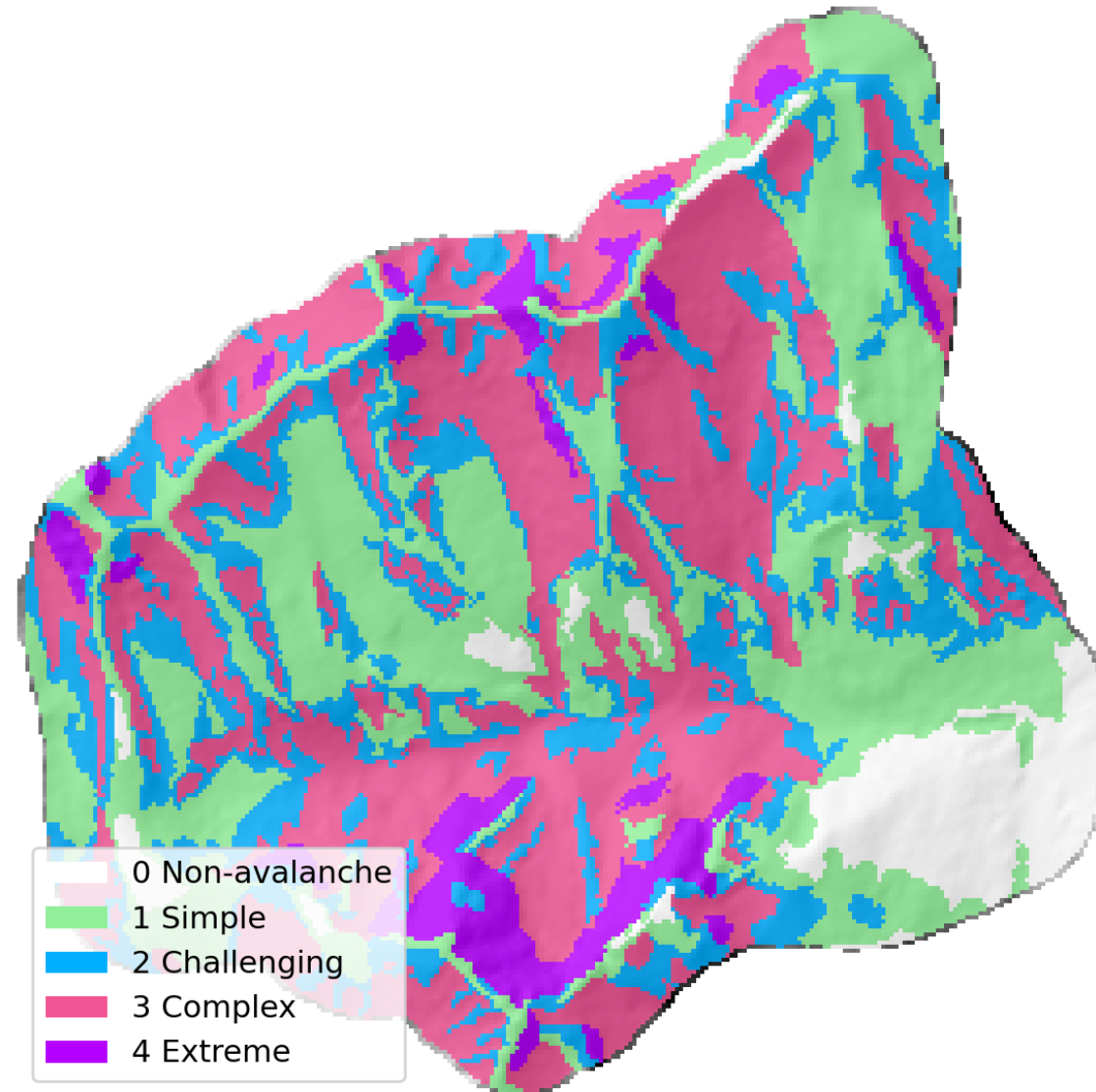
Helpers from 08:15, with USB copies. Please download the folders before Saturday if you can — conference wifi is often slow.

Five steps. Typical and infrequent stay split.



One parameterization cannot stand in for everyday events and the larger, less frequent avalanches that still belong on an ATES map.

Connaught Creek, Rogers Pass — 13 km²

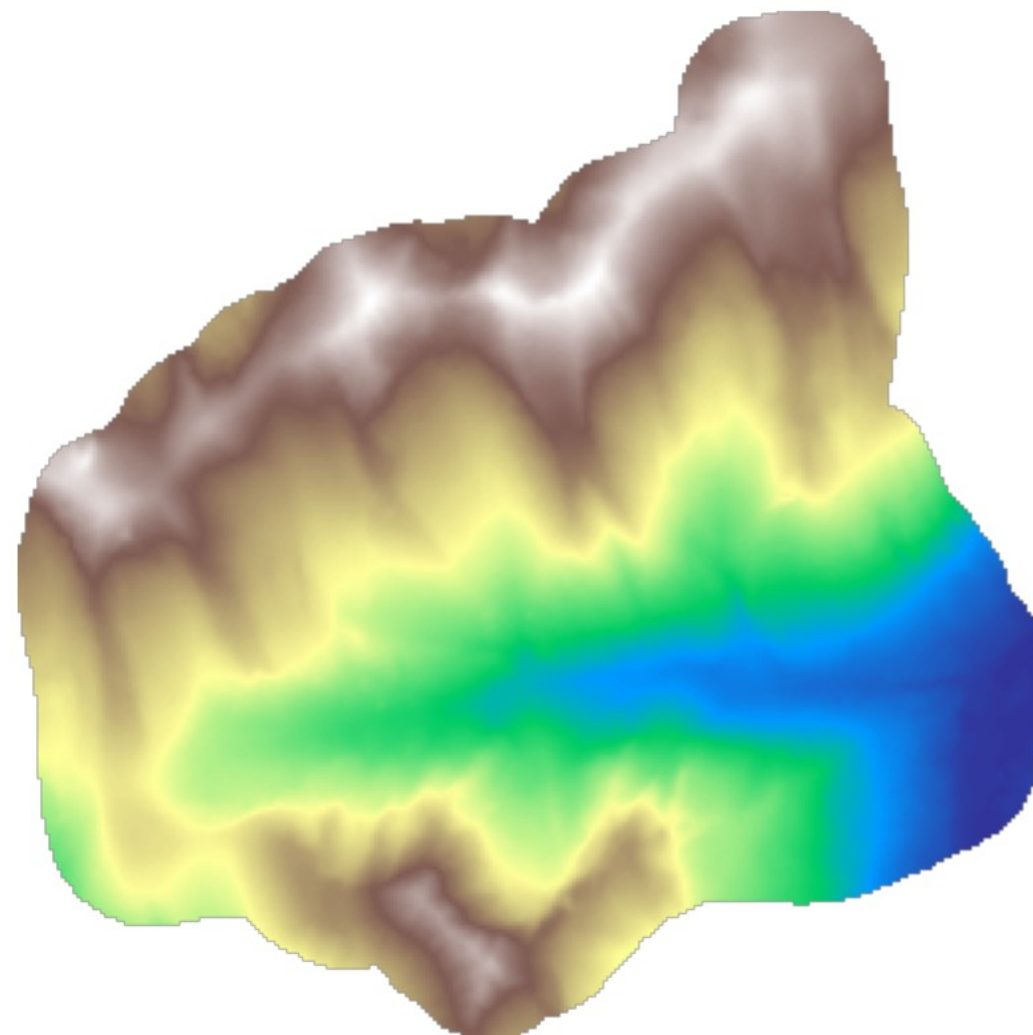


30 m surface. Canopy-top in forest. Slope is smoothed.

Hillshade



ALOS AW3D30 (m)



Summer greenness. Winter conifer, deciduous, and gully paths.

Summer NDVI



Winter NDSI



Binary conifer



PRA uses canopy cover. The classifier uses opening size.

Canopy cover % (3×3)

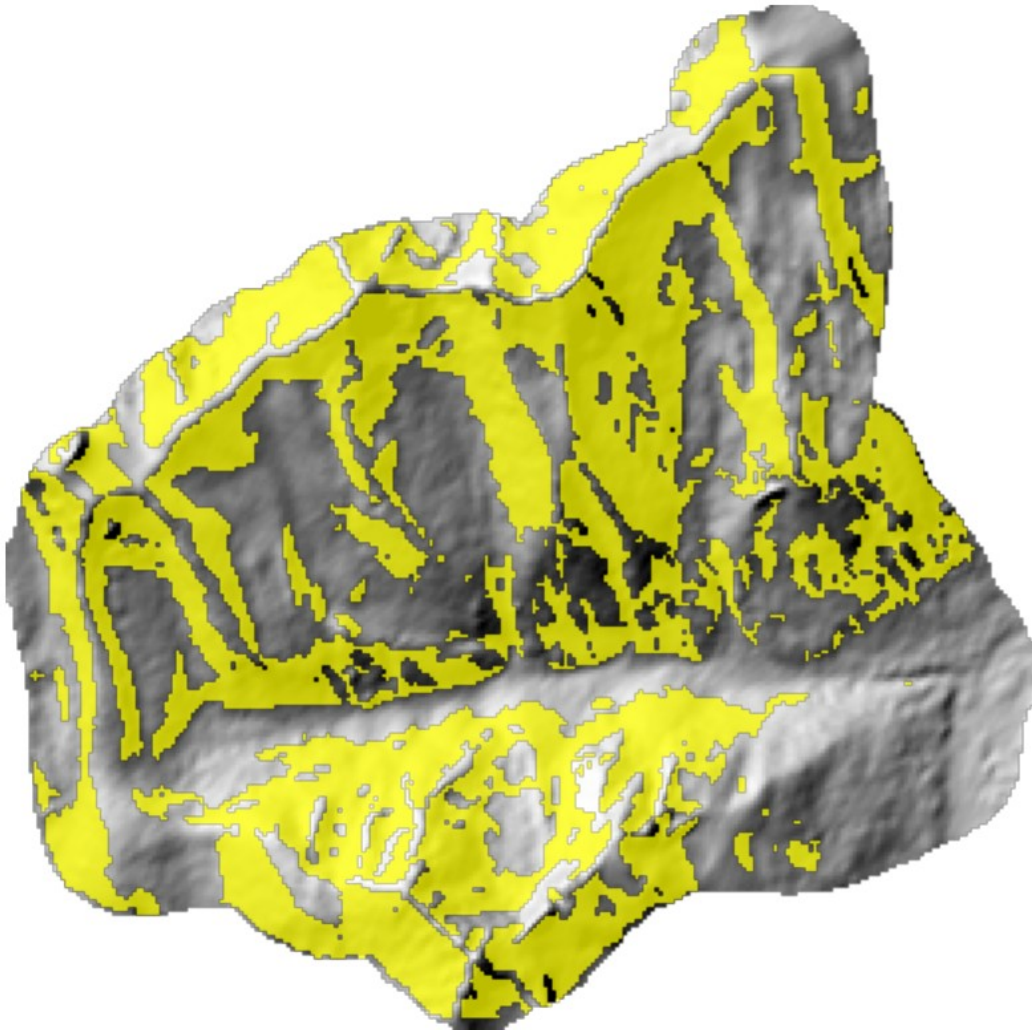


Gap area (ha)

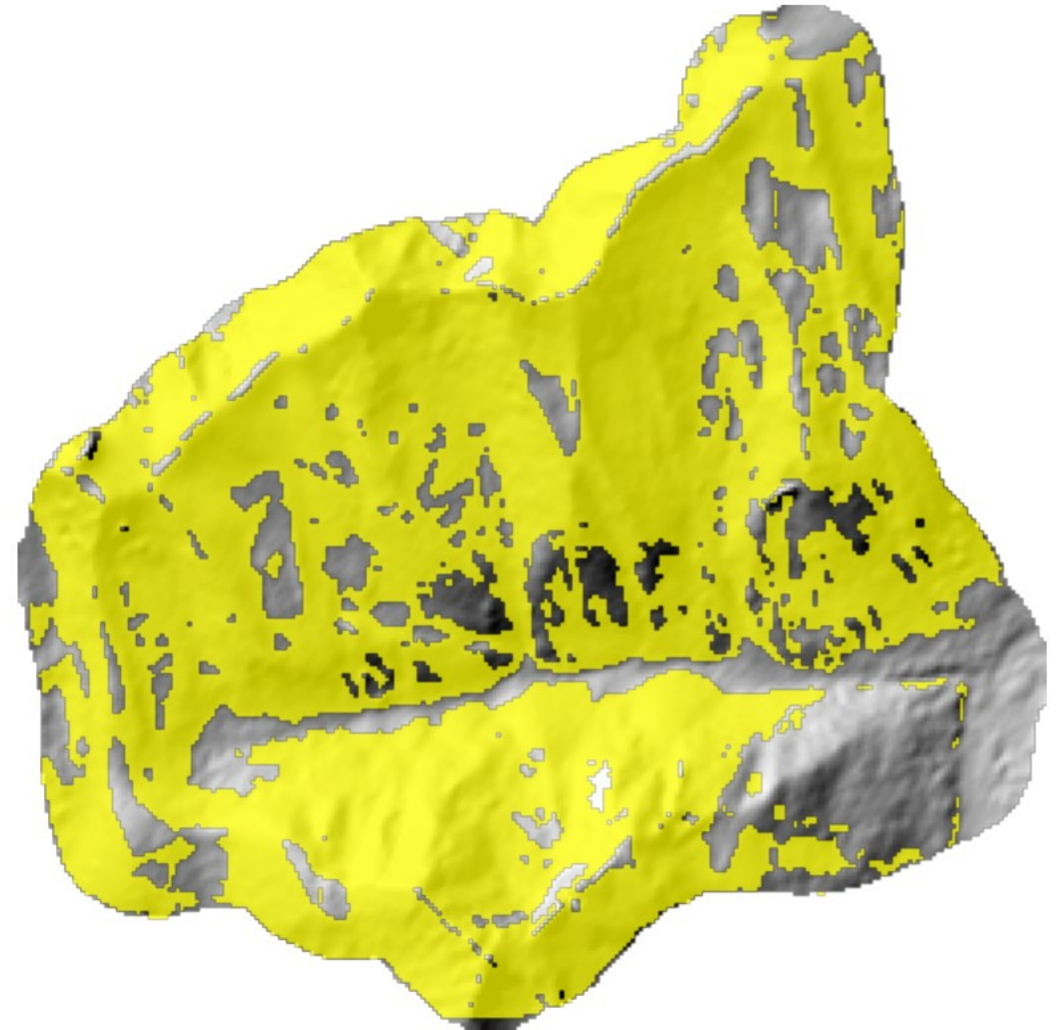


Typical is tighter. Infrequent picks up start zones below 30°.

Typical PRA

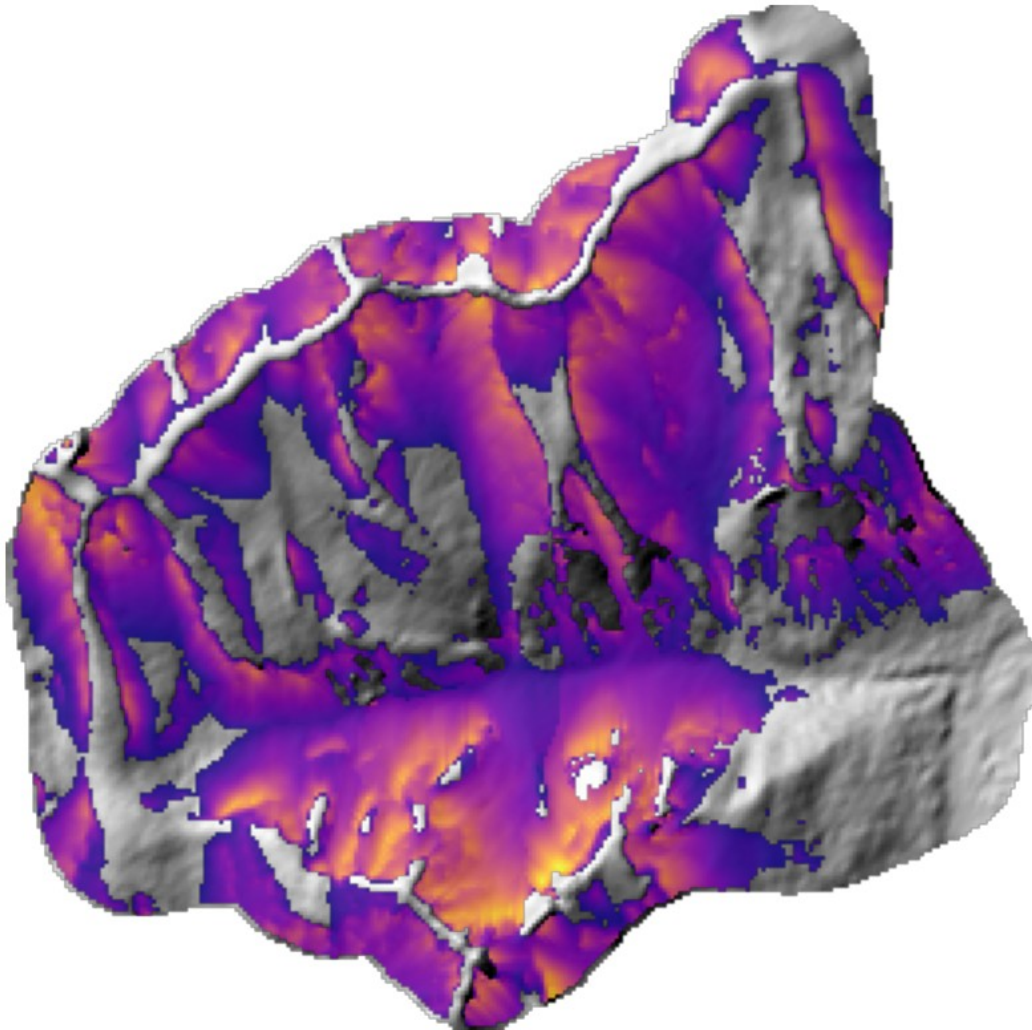


Infrequent PRA

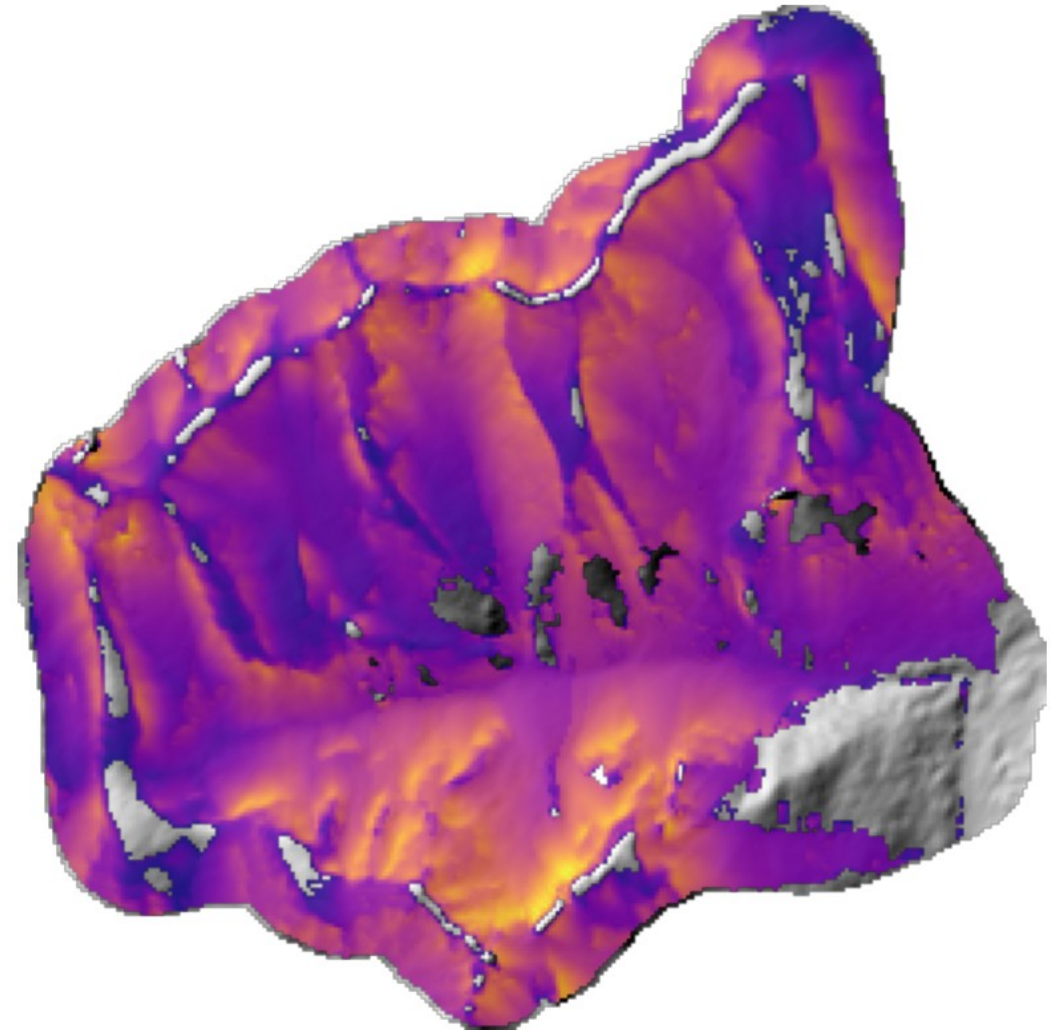


com4FlowPy — typical α 30°, infrequent α 18°. Not RAMMS.

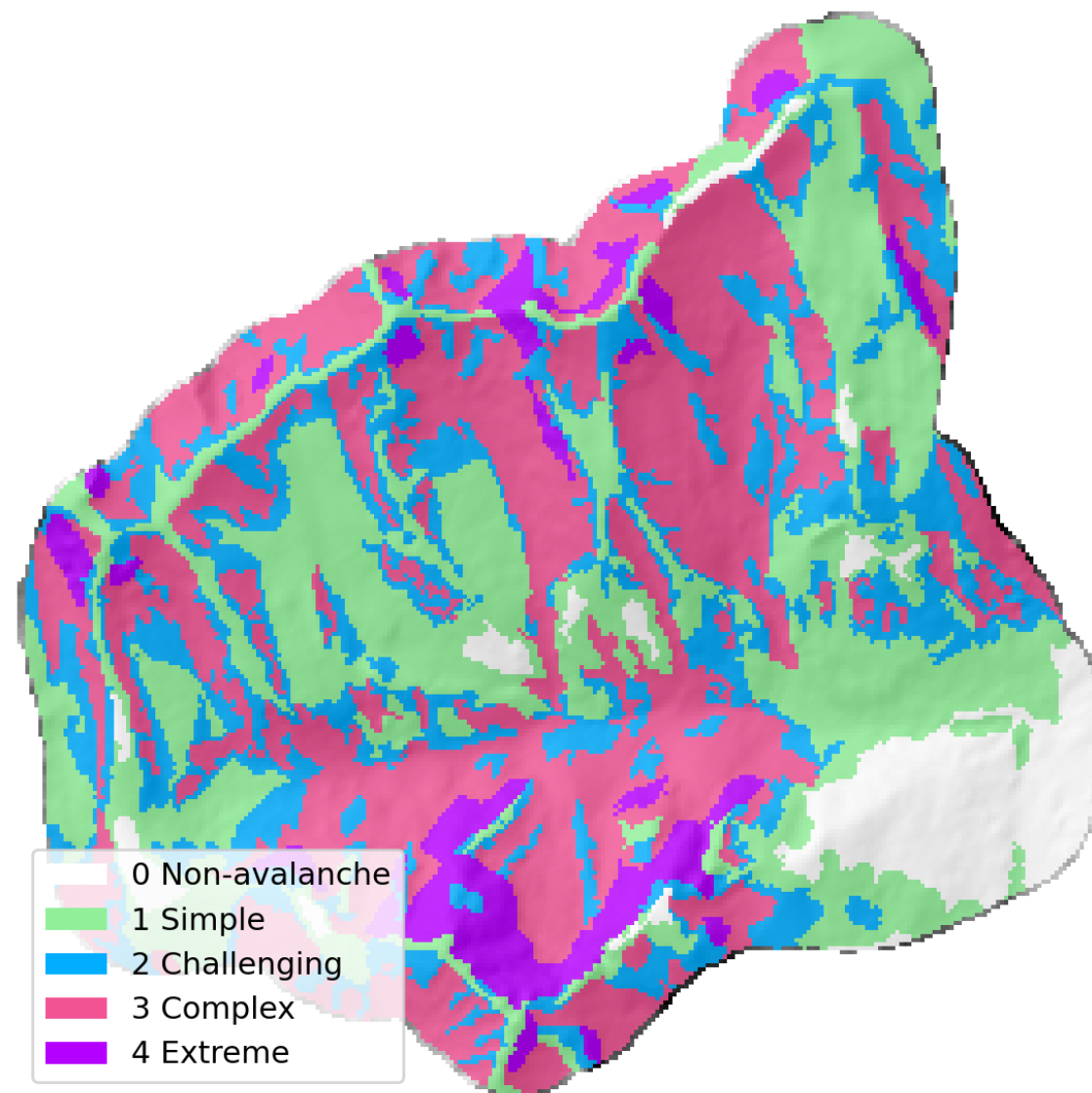
Typical reach · α 30°



Infrequent reach · α 18°



Infrequent sets the envelope. A floor cannot be averaged down.



Floors first. Then the rest is averaged.

Simple	Infrequent runout reach
Challenging	Typical runout reach
Complex	Typical overhead / contributing area
Extreme	Slope > 45° and open canopy

TAKE-HOME

Point this at a home range with an AOI, a 30 m surface, and a canopy raster.

Keep typical / infrequent parameters unless you have a reason and a validation set.

A map from this tutorial is not an operational ATES product.

Repo, pins, and papers: SETUP.md and CITING.md in the workshop repository.

Best-practices document is in development.